

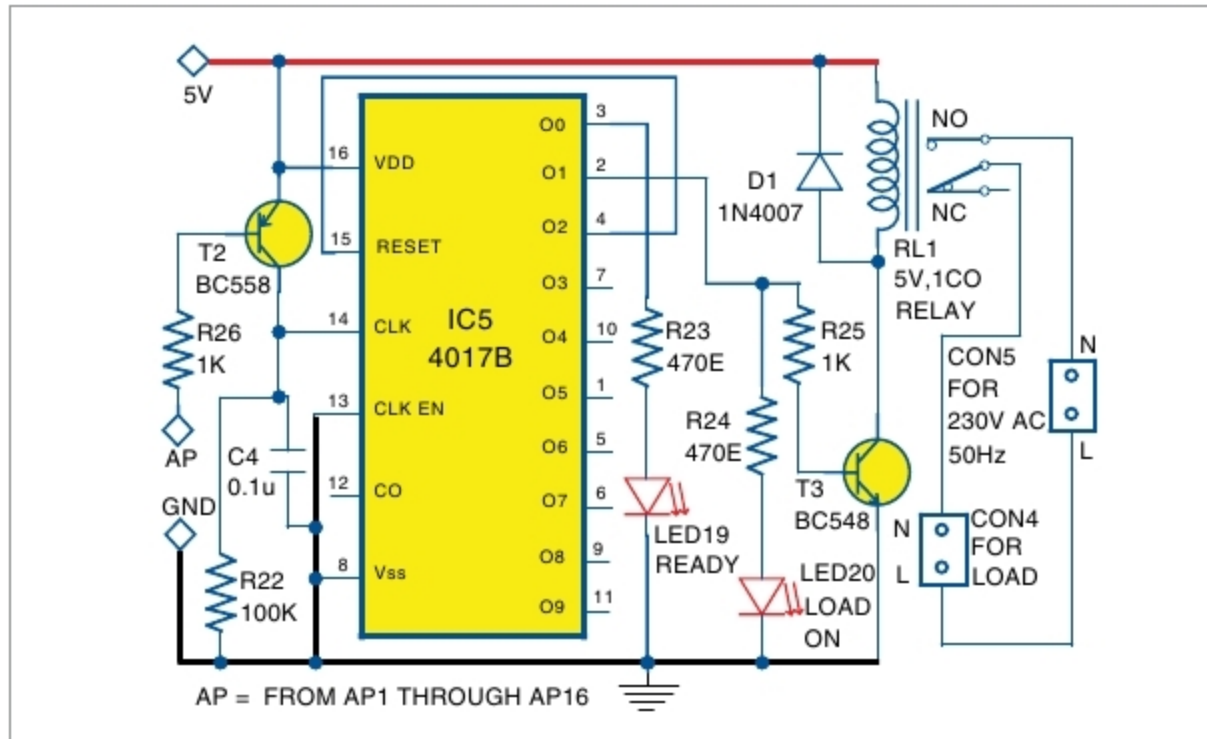


Fig. 3: Circuit diagram of bistable section

FUNCTIONS OF DIFFERENT LEDs

LED	Function
LED1	Blinks when a key is pressed on keypad
LED18	Blinks when valid data is received
LED19	Glow when circuit is ready to control connected appliance
LED20	Glow when connected appliance is activated
LED2-LED17	Glow when a respective load (AP1-AP16) is on

bistable circuit that will convert the single-pulse signal into two stable states (high and low) to drive a relay.

Bistable section. The circuit diagram of the bistable section is shown in Fig. 3. It is built around decade counter 4017B (IC5), 5V relay (RL1) and a few other components. Here, 4017B is used as a bistable multivibrator. Output state of the bistable does not change until the next input is received. AP is the appliance control input to IC5. It

could be any one of the inputs from AP1 through AP16, from IC4. Connect the appliance to be controlled across the contacts of the relay.

Working of the circuit is simple. When you press a key on the keypad, the input signal is encoded and transmitted through the RF transmitter module. The receiver receives the serial data and converts it into parallel data (BCD). This BCD data is available across output pins of IC4.

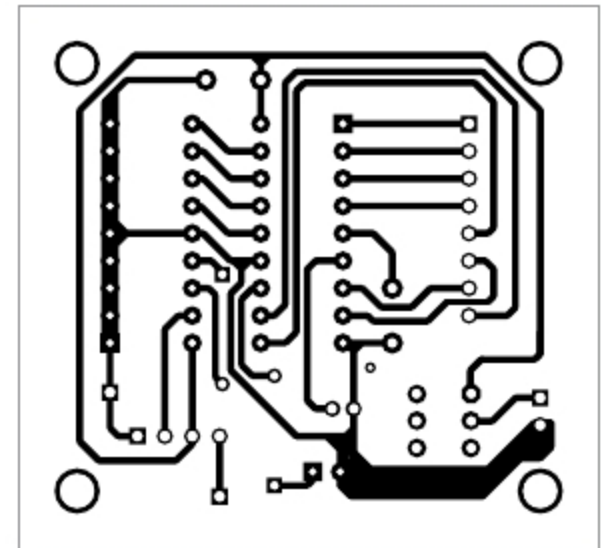


Fig. 4: Actual-size PCB layout of transmitter circuit

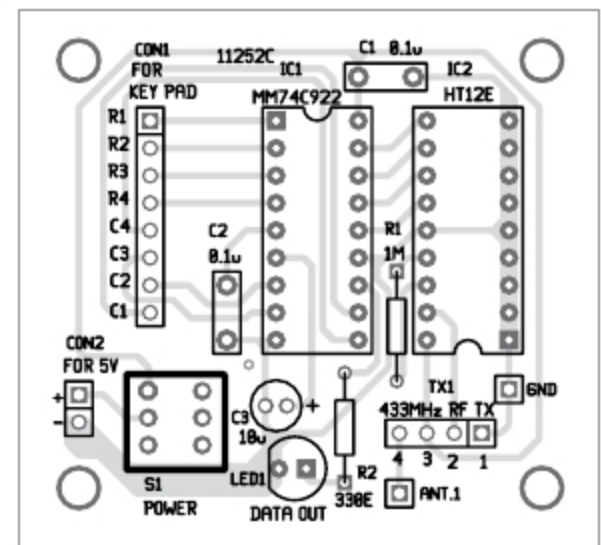


Fig. 5: Components layout for the PCB shown in Fig. 4

At each output pin of IC4 a bistable circuit is connected, which converts the signal into two stable states depending on the status of the keypad. For example, if input key 1 (not shown here) is pressed, AP1 will be low and RL1 will be energised. The load or appliance connected at CON4 will be on at the same time LED17 and LED20 will be on. And, when input key 0 or any other key is pressed again, AP1

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